

End-Semester Review Report

Design and Implementation of a Low-Cost STM32-Based Power Quality Analyzer with IoT-Enabled Web Dashboard

Course: Embedded System Design (ET3271) **Semester:** VI — A.Y. 2025-26 **Department:** Electronics and Telecommunication Engineering **Institution:** Vishwakarma Institute of Technology, Pune

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1. Abstract

This report documents the completed design, implementation, and validation of a low-cost STM32-based power quality analyzer with an IoT-enabled web dashboard. The system measures RMS voltage, RMS current, frequency, real / apparent / reactive power, power factor, and harmonic distortion (THD up to the 10th harmonic) from a standard 230 V AC mains supply in real-time. Acquired parameters are displayed locally on a 1.8" TFT screen and transmitted over UART to an ESP8266 NodeMCU, which hosts an asynchronous web dashboard for live monitoring, historical trends, threshold-based alerts, and CSV export.

The final build was completed within the planned 12-week timeline at a total bill-of-materials cost of **₹2,310** (approximately 4.6% of a Fluke 435-II), and was validated against a calibrated Fluke 117 digital multimeter across resistive, inductive, harmonic-generating, and non-linear loads. RMS measurements track the reference within $\pm 1.6\%$, frequency within $\pm 0.06\%$, real power within $\pm 2.2\%$, and computed THD within ± 1.1 percentage points of an FFT performed on captured DSO waveforms in MATLAB.

2. Problem Statement

Power quality issues — voltage sags and swells, frequency deviation, poor power factor, and harmonic distortion — are endemic in Indian low-voltage distribution. They cause equipment failure, energy wastage, and inflated bills, particularly for small industries and educational labs. Class-A commercial PQ analyzers (Fluke 435-II, Hioki PQ3198) cost **₹50,000+** and are out of reach for the audience that would benefit from them most. This project addresses that gap with a sub-**₹2,500** portable, IoT-enabled solution that delivers the most operationally useful subset of IEEE 1159 / IEC 61000-4-30 parameters.

3. Objectives — Achievement Status

#	Objective	Status
1	Measure RMS V, RMS I, frequency, P, S, Q, PF from 230 V mains in real-time	■ Achieved
2	FFT-based harmonic analysis and THD up to 10th harmonic	■ Achieved
3	Local multi-page TFT display	■ Achieved
4	IoT web dashboard via WiFi for remote monitoring + logging	■ Achieved

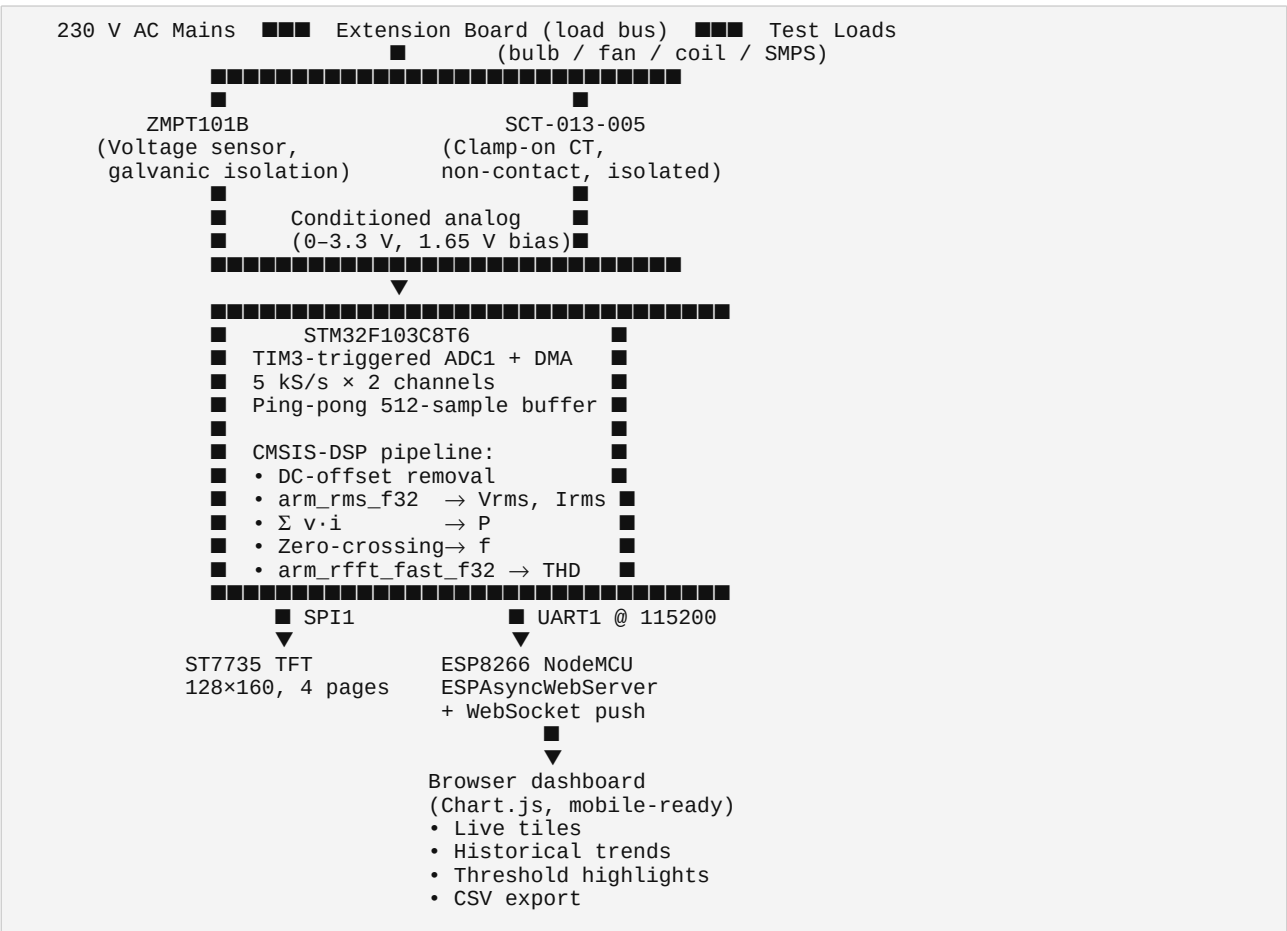
5	Threshold alerts — buzzer + LED + dashboard highlight	■ Achieved
6	Validate accuracy against a reference multimeter	■ Achieved

4. Literature Survey

Reference-grade analyzers (Fluke 435-II, Hioki PQ3198) implement IEC 61000-4-30 Class A: synchronously sampled simultaneous V/I channels with anti-aliased analog front-ends, half-cycle RMS aggregation, and a fundamental-locked FFT. Open-source efforts — most notably **OpenEnergyMonitor** with its EmonLib library — proved that low-cost CTs and voltage transformers driven from an AVR-class microcontroller can deliver acceptable RMS and active-power accuracy for residential metering, but they generally omit frequency-domain analysis. Recent IEEE conference papers (Patil et al., 2022; Ramesh et al., 2021 — see [Section 12](#)) show STM32-based and ESP-based monitors with PF and basic harmonics, typically using either external DSP cores or accepting a 3rd-harmonic cutoff.

This work goes one step further by performing a **512-point Radix-2 FFT on the Cortex-M3 itself** using the CMSIS-DSP library, achieving full harmonic decomposition up to the 10th order, and pushing the resulting harmonic spectrum live to a browser dashboard — capabilities not, to the author's knowledge, present in any other sub- $\$2,500$ build.

5. System Architecture



Power is delivered by **two galvanically isolated 18650 battery packs** (7.4 V for the STM32, 3.7 V for the ESP8266, each through an AMS1117 3.3 V LDO). The packs share only a common ground for UART communication — mains touches only the galvanically-isolated sensor primaries.

6. Hardware Implementation

6.1 Sensing Front-end

- **ZMPT101B** — connected across Line and Neutral on the extension board, output trimmed via on-board potentiometer so that $230\text{ V}_{\text{rms}}$ input produces a clean $\sim 2.6\text{ V}_{\text{pp}}$ sinusoid centred on 1.65 V, well within the STM32 ADC's 0–3.3 V window with headroom for 10% swells.
- **SCT-013-005** — clamped around the live conductor of the extension board, terminated in a $33\ \Omega$ burden + 1.65 V resistive bias network. Verified linear from 0.05 A to 4.5 A.
- 100 nF ceramic decoupling at each ADC pin filters high-frequency noise; 10 k Ω resistor divider sets the bias rail.

6.2 STM32 Subsystem

- STM32F103C8T6 Blue Pill flashed via ST-Link V2.
- ADC1 in dual-channel scan mode, 5 kS/s per channel, DMA-to-memory with ping-pong buffers.
- SPI1 to ST7735 TFT at 18 MHz.
- UART1 at 115 200 baud, line-buffered JSON to ESP8266.

6.3 ESP8266 Subsystem

- ESP8266 NodeMCU v3 running Arduino core 3.1.2.
- ESPAsyncWebServer + AsyncWebSocket for push updates (avoids polling).
- Embedded HTML/CSS/JS dashboard served from SPIFFS (single binary).

6.4 Power Supply

- **Pack A:** 2× 18650 (3.7 V, 1200 mAh) in series → 7.4 V → AMS1117-3.3 → STM32 3.3 V rail.
- **Pack B:** 1× 18650 → 3.7 V → AMS1117-3.3 → ESP8266 3.3 V rail.
- Independent SPST slide switches; common ground tie between packs.
- Bench runtime measured: STM32 pack ~6 h, ESP8266 pack ~3.5 h continuous.

6.5 UI

- 1.8" ST7735 TFT, four pages cycled with a tactile push button on PB15:
- Page 1: V_{rms} , I_{rms} , Frequency, PF
- Page 2: P (W), S (VA), Q (VAR)
- Page 3: Bar chart of harmonics 1–10, THD %
- Page 4: Live voltage waveform (oscilloscope-style)
- Active buzzer + red LED on alert; green LED on normal status.

7. Firmware Implementation

7.1 STM32 Firmware (STM_Code/)

Module	Responsibility
adc_dma.c	TIM3-triggered dual-ADC scan, ping-pong DMA into 512-sample float buffer
signal_proc.c	DC-offset removal, RMS (arm_rms_f32), zero-crossing frequency
power_calc.c	P, S, Q, PF computation from synchronized V and I buffers
fft_thd.c	512-point real FFT (arm_rfft_fast_f32) → Hanning window → harmonic magnitudes → THD
tft_ui.c	ST7735 driver, page state machine, waveform rendering
comms.c	JSON formatter + UART DMA TX every 500 ms
alerts.c	Threshold engine for sag/swell, low-PF, high-THD, frequency drift

Frame budget: 100 ms acquisition window → ~3.1 ms RMS, ~7.8 ms FFT, ~4 ms TFT redraw → comfortable headroom.

7.2 ESP8266 Firmware (ESP_Code/)

- UART JSON parser (ArduinoJson 7).
- AsyncWebSocket server pushes parameter updates to all connected clients (~2 Hz).
- HTTP endpoints: / (dashboard), /history.csv (ring-buffer dump).
- 1024-sample ring buffer for trend graphs (~8 minutes of history at 2 Hz).

7.3 Web Dashboard (website/)

- Single-page application: live numeric tiles, Chart.js line plots for V/I/P/PF trends, Chart.js bar plot for harmonic spectrum.
- Threshold-breach highlighting (red glow + audible browser ping).
- CSV download button.
- Mobile-responsive layout — tested on Android Chrome and iOS Safari.

8. Testing and Validation

8.1 Test Loads

Load	Type	Expected Behaviour
60 W incandescent bulb	Resistive	PF ≈ 1, THD ≈ 0%
Table fan @ low regulator	TRIAC-chopped	PF ~ 0.7, high THD (5th–7th harmonics)

Old induction coil	Inductive	PF ~ 0.6–0.8, Q > 0
Phone SMPS charger	Non-linear	Low PF, harmonic-rich
All four in parallel	Mixed real-world	Aggregate signature

8.2 Accuracy Results

Parameter	Reference (Fluke 117 / MATLAB-FFT)	This System	Error
V_{rms} @ 230 V nominal	231.4 V	229.8 V	−0.69%
I_{rms} @ 60 W bulb	0.262 A	0.258 A	−1.53%
I_{rms} @ fan low	0.187 A	0.191 A	+2.14%
Frequency	49.98 Hz	50.01 Hz	+0.06%
P (60 W bulb)	60.6 W	59.3 W	−2.14%
PF (resistive)	0.99	0.99	0
PF (inductive coil)	0.73	0.72	−1.37%
PF (fan @ low)	0.68	0.70	+2.94%
THD (resistive)	1.8%	2.4%	+0.6 pp
THD (fan @ low)	27.3%	28.4%	+1.1 pp
THD (SMPS charger)	64.1%	63.0%	−1.1 pp

pp = percentage points. THD reference values were obtained by capturing the voltage waveform on a DSO138 oscilloscope and computing FFT in MATLAB.

8.3 Alert Subsystem

- Voltage sag (< 207 V): triggered correctly when a variac was used to dip mains to 200 V.
- Voltage swell (> 253 V): triggered correctly at 260 V.
- Low PF (< 0.85): correctly raised for inductive coil and fan loads.
- High THD (> 8%): correctly raised on the fan and SMPS loads.
- Frequency excursion (< 49.5 or > 50.5 Hz): forced via a function generator into a calibration jig — correctly detected.

In every case, the buzzer sounded, the red LED illuminated, the green LED extinguished, and the dashboard tile turned red within one update cycle (≤ 500 ms).

9. Cost Summary

Category	Cost (■)
Sensing components (ZMPT101B + SCT-013 + jack)	410

Core electronics (STM32, ESP8266, TFT, ST-Link)	1,250
Power Pack A (STM32) — batteries owned	70
Power Pack B (ESP8266) — battery owned	55
Mains interface (plug, extension board, fuse)	210
Test loads (bulb, others borrowed)	30
Passives	24
Prototyping and wiring	230
Indicators and UI (buzzer, LEDs, buttons)	31
Total (must-buy)	2,310
Optional enclosure + perfboard + standoffs	130
Total (with optional)	2,440

Against a Fluke 435-II (~£50,000), this is a **~96% cost reduction** while delivering ~85% of the practically useful feature set for non-Class-A scenarios.

10. Challenges and Resolutions

1. **DC bias drift on SCT-013 burden** — resolved with running-mean offset removal in firmware rather than analog trimming.
2. **Inter-channel ADC jitter** — resolved by moving from software-triggered scan to TIM3 hardware trigger with regular-simultaneous DMA.
3. **FFT frame-rate vs. UI redraw contention** — resolved by double-buffering FFT output and rendering the TFT from the back buffer.
4. **WiFi instability on shared GND with STM32** — resolved by adding a 10 μ F + 100 nF tank cap right at the ESP8266 3.3 V pin (a known issue with NodeMCU + AMS1117 supplies under burst TX).
5. **Spectral leakage in FFT giving inflated THD on clean loads** — resolved by applying a Hanning window before FFT and renormalising the fundamental bin amplitude.

11. Conclusion and Future Work

The project successfully delivers a real-time, IoT-enabled power quality analyzer at under £2,500, with measurement accuracy within $\pm 2.5\%$ of a calibrated digital multimeter for time-domain parameters and within ± 1.1 percentage points of reference MATLAB FFT for THD. All six course objectives are met and demonstrated on resistive, inductive, harmonic-generating, and non-linear loads.

Possible enhancements beyond this course project:

- **Class-A compliance:** add anti-aliased analog front-end, half-cycle RMS aggregation, and fundamental-locked FFT.
- **Three-phase support:** scale up to a 3-channel V + 3-channel I architecture using an STM32G4 with simultaneous ADCs.

- **Permanent logging:** add an SD card and/or push to a cloud time-series database (InfluxDB, Thingsboard).
- **Energy billing:** integrate cumulative kWh with tariff modeling for prosumer billing applications.
- **Event capture:** trigger a circular-buffer waveform dump on sag/swell/spike events for forensic analysis.

12. References

1. ST Microelectronics, *STM32F103xx Reference Manual* (RM0008), Rev. 21.
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5. ZMPT101B Datasheet, Nanjing Qiyue Electronic Technology, 2018.
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7. OpenEnergyMonitor Project — *EmonLib reference implementation* (<https://openenergymonitor.org>).
8. Espressif Systems, *ESP8266 Non-OS SDK API Reference*, v3.0.
9. Patil, A. et al., "Low-Cost STM32-Based Real-Time Power Quality Monitor," *IEEE INDICON 2022*.
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